

AGATA, Russian Federation

WMO: 233830

Lat: 66.926N

Lon: 93.493E

Elev: 913

StdP: 14.22

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-61.2	-57.2	-64.3	0.1	-61.6	-60.6	0.1	-56.9	18.7	-1.0	16.6	1.9	0.0	30	0.816

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	77.4	61.3	72.3	58.3	68.2	56.6	62.6	73.4	60.4	69.5	58.1	66.1	4.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	76.5	65.5	56.5	70.2	63.5	54.2	64.6	61.8	28.7	74.1	27.2	69.8	25.6	66.0	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	15.1	13.1	-61.7	82.8	4.6	4.4	-65.0	85.9	-67.7	88.4	-70.3	90.9	-73.7	94.1		
			-61.4	66.6	5.0	2.3	-65.0	68.3	-67.9	69.7	-70.7	71.0	-74.4	72.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	-30.2	-24.6	-3.3	14.0	32.0	48.6	57.5	50.6	38.2	20.9	-11.7	-26.6		
	DBStd	33.71	17.86	16.26	16.95	12.99	9.18	7.85	6.01	6.00	7.43	12.31	19.94	18.96		
	HDD50	13539	2485	2088	1651	1079	558	121	8	66	357	901	1851	2374		
	HDD65	18618	2950	2508	2116	1529	1022	493	243	448	803	1366	2301	2839		
	CDD50	409	0	0	0	0	1	79	240	85	4	0	0	0		
	CDD65	12	0	0	0	0	0	1	10	1	0	0	0	0		
	CDH74	271	0	0	0	0	1	37	213	20	0	0	0	0		
	CDH80	49	0	0	0	0	0	5	43	1	0	0	0	0		
	WSAvg	3.6	1.4	1.7	3.2	5.1	4.9	3.8	3.9	4.2	5.3	5.6	2.5	1.2		
	PrecAvg	18.3	1.0	1.0	0.7	0.8	1.1	2.2	2.4	2.6	2.1	1.8	1.4	1.2		
(15) Precipitation	PrecMax	25.2	2.4	10.4	1.7	2.7	2.4	4.7	6.8	4.4	5.1	3.5	2.9	2.4		
	PrecMin	10.3	0.3	0.1	0.1	0.1	0.0	0.1	0.0	0.4	0.6	0.6	0.2	0.7		
	PrecStd	3.7	0.5	1.6	0.4	0.6	0.6	1.1	1.3	1.1	0.9	0.7	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.3	21.0	35.2	45.5	62.6	79.8	84.8	77.0	59.8	47.5	34.8	22.0		
		MCWB	18.3	19.1	30.4	37.2	48.4	60.9	63.1	63.5	51.6	42.9	32.3	20.1		
	2%	DB	11.8	14.5	30.6	41.3	53.7	73.2	79.8	69.9	55.0	41.9	29.6	15.4		
		MCWB	10.4	12.9	27.4	35.0	43.8	57.3	61.6	57.4	48.1	39.2	27.6	14.0		
	5%	DB	4.3	7.3	25.8	37.8	48.9	68.7	75.3	65.5	51.9	38.1	23.4	8.0		
		MCWB	3.2	5.9	23.2	34.0	40.6	54.2	60.6	56.3	46.6	35.6	21.7	7.1		
	10%	DB	-2.5	0.6	21.8	34.1	44.5	64.3	71.0	61.7	48.8	35.7	16.1	2.1		
		MCWB	-3.1	-0.4	19.5	30.8	37.9	52.5	58.6	54.3	44.5	33.3	14.6	1.3		
	0.4%	WB	18.6	19.0	30.5	38.5	49.0	62.9	66.0	63.4	54.9	43.2	32.4	21.0		
		MCDB	20.5	21.0	34.0	43.5	60.4	76.9	77.5	73.5	58.9	46.5	34.8	22.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	10.4	12.8	27.1	36.0	44.7	58.9	63.7	59.5	49.8	39.1	27.6	13.8		
		MCDB	11.8	14.4	30.3	39.7	52.7	69.0	74.5	66.6	52.3	41.6	29.6	15.3		
	5%	WB	3.3	6.1	23.4	33.9	41.3	56.1	62.0	57.1	47.4	35.9	21.9	7.1		
		MCDB	4.4	7.4	25.9	38.0	47.4	65.5	72.2	63.6	50.9	37.7	23.6	7.8		
	10%	WB	-3.2	-0.4	19.5	30.6	38.6	53.4	60.2	55.2	44.7	33.3	14.7	1.4		
		MCDB	-2.5	0.6	21.9	34.2	43.9	62.8	69.2	60.6	48.5	35.5	16.0	2.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	17.1	22.2	21.9	15.3	18.1	18.9	15.7	11.5	8.9	14.0	14.3		
		MCDBR	17.7	19.5	16.2	17.6	20.8	29.5	30.2	25.7	17.9	7.8	10.4	17.0		
		MCWBR	17.1	18.6	14.4	13.7	12.6	16.0	14.7	14.9	12.3	6.8	9.8	16.1		
		MCDBR	17.8	19.4	15.8	16.1	18.9	25.2	24.1	20.7	14.4	7.1	10.3	17.0		
	5% WB	MCWBR	17.1	18.5	14.1	12.9	12.1	15.3	13.1	13.0	10.9	6.4	9.8	16.1		
(40) Clear-Sky Solar Irradiance	taub		0.209	0.257	0.303	0.330	0.349	0.366	0.414	0.401	0.324	0.255	0.191	N/A		
	taud		1.842	1.982	2.004	2.027	2.101	2.355	2.247	2.236	2.371	2.093	1.877	N/A		
	Ebn at Noon		43	173	228	257	266	266	245	233	230	174	47	0		
	Edh at Noon		5	18	31	40	41	33	36	33	22	16	4	0		
(44) All-Sky Solar Radiation	RadAvg		16	198	689	1377	1848	1751	1530	947	537	205	41	0		
	RadStd		3	15	64	107	130	191	180	107	88	38	6	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page